

Depth, Time, and State: Three Reductions in Autoregressive Decoding

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Abstract

Three recurring complaints about autoregressive language models — that speculative decoding needs an auxiliary draft network, that reasoning is externalised into emitted tokens, and that decoder-only transformers are elaborate recurrent networks — are usually treated separately. We show they are three instances of one question: along which axis is a model permitted to spend *serial* computation, and how does it carry state while doing so. We formalise each along its own axis (depth, time, state) and prove what the informal complaint does and does not license. For depth, we show speculative sampling is exact for an arbitrary draft distribution, so correctness imposes no constraint on the draft; we give a cost model interpolating between memory- and compute-bound decoding under which self-drafting by early exit dominates any external draft of equal relative cost, and lower-bound the acceptance rate by the residual tail energy. For time, we show the strong form of the latent-reasoning complaint is false under standard complexity assumptions — serial compute is load-bearing — but that a counting argument separates the *media*: a chain-of-thought step transports at most $\log_2 |\mathcal{V}|$ bits, a latent step transports the state dimension. Discretisation, not serial computation, is the object of the complaint. For state, we show the transformer’s recurrent form is real but nearly vacuous, that a constrained gated linear recurrence coincides exactly with the HMM forward algorithm, and — our main negative result — that *diagonally*-gated linear attention, the form used by most modern sub-quadratic architectures, provably cannot realise a mixing HMM transition within a single layer. We close with a capacity lower bound that fixes what unbounded state is buying. All claims are stated to be falsifiable, and we give a protocol at $\leq 1\text{B}$ parameters with pre-registered kill criteria.

1 Introduction

Three complaints about contemporary language models recur often enough to be folklore. Speculative decoding [1, 2] accelerates inference by having a second, smaller network guess ahead — why can a model not guess ahead using its own early layers? Reasoning is elicited by having models emit long chains of intermediate tokens — does that not indicate a model that never learned to reason internally? And decoder-only transformers, whose key-value cache is carried forward and updated at every step, are recurrent networks with an unusually large state — so why not build the recurrent network directly?

Each complaint is usually argued on its own. We observe that they are the same question asked along three different axes: *where is a model permitted to spend serial computation, and what carries state while it does so?* Speculative decoding is about the *depth* axis, where the residual stream already contains a usable intermediate answer that is discarded. Chain of thought is about the *time* axis, where serial computation is routed through the vocabulary. The recurrent-form question is about the *state* axis, where an unbounded cache substitutes for a learned compression.

Our contribution is not a new architecture but a precise accounting of what each complaint does and does not license. We find the three fare very differently under formalisation, and in two cases the honest conclusion is not the one the complaint anticipates.

Depth. Speculative sampling is exact for an *arbitrary* draft distribution (Theorem 3.1), so correctness never justified a separate draft network; the question is entirely one of acceptance against cost. We give a cost model interpolating between memory- and compute-bound decoding (Definition 2.4) under which early-exit self-drafting strictly dominates any external draft of comparable relative cost, in every regime (Theorem 3.20), and lower-bound the acceptance rate by the residual tail energy (Theorem 3.7). We also identify a cost term that the informal argument omits and that materially changes the conclusion for small models: every draft step must stream the unembedding, which caps the attainable speedup (Corollary 3.22) unless the draft head is shortlisted (Corollary 3.24).

Time. The strong form of the complaint is false: under $\text{TC}^0 \neq \text{P}$, serial steps buy computational power that fixed depth cannot recover (Theorem 4.1). The comparison between media is a genuine trade-off, and it runs *opposite* to the usual claim. A chain-of-thought step writes only $\log_2 V \approx 17$ bits, but those writes accumulate; a fixed-size latent loop writes far more per step but overwrites. Past a computable crossover — about 238 steps for a 1024-dimensional model — the token stream is the larger memory (Corollary 4.5). The defensible version of the complaint is therefore that latent reasoning should *append* rather than overwrite.

State. The recurrent form is real but nearly vacuous as usually stated; its content is incrementality, not existence (Theorem 5.2). A constrained dense recurrence does coincide exactly with the HMM forward algorithm (Theorem 5.6). Our main negative result is that this is *not* a statement about gated linear attention: we prove that no injective linear embedding carries the forward algorithm into a diagonally-gated linear-attention recurrence (Theorem 5.8), by a support-monotonicity argument and by simultaneous diagonalisability. Since diagonality is precisely what makes these architectures parallelisable, “constrain a gated RNN and obtain HMM-like decoding” is false as stated.

Falsifiability. Every claim above is either proved, refuted, or labelled open. Section 8 gives an experimental protocol at $\leq 1\text{B}$ parameters in which each thesis has a pre-registered kill criterion, and two of the four experiments require no training.

2 Preliminaries

2.1 Residual-stream language models

Definition 2.1 (Residual-stream language model). A *residual-stream language model* over a finite vocabulary $\mathcal{V}$ is a tuple $M = (E, (F_\ell)_{\ell=1}^L, N, W_U)$ with embedding $E : \mathcal{V} \rightarrow \mathbb{R}^d$, causal blocks $F_\ell : (\mathbb{R}^d)^* \rightarrow \mathbb{R}^d$, a normalisation map $N : \mathbb{R}^d \rightarrow \mathbb{R}^d$, and unembedding $W_U \in \mathbb{R}^{|\mathcal{V}| \times d}$. On a prefix $x = x_1 \dots x_T \in \mathcal{V}^*$ the model computes, for $t \leq T$,

$$h_t^{(0)} = E(x_t) + \text{pos}(t), \quad h_t^{(\ell)} = h_t^{(\ell-1)} + F_\ell(h_{1:t}^{(\ell-1)}), \quad \ell = 1, \dots, L, \quad (1)$$

where F_ℓ depends on positions $\leq t$ only (causality). Its output distribution is $p_M(\cdot | x) = \text{softmax}(W_U N(h_T^{(L)}))$.

We write L for depth, d for width, $V = |\mathcal{V}|$, and suppress the position subscript when it is the final one. Throughout, $\text{TV}(p, q) = \frac{1}{2} \|p - q\|_1$ and $\|W\|_{2,\infty} = \max_i \|W_{i,:}\|_2$ is the largest row norm.

Definition 2.2 (Layer- ℓ readout). For $\ell \in \{0, \dots, L\}$ the *layer- ℓ readout* of M is

$$p_\ell(\cdot | x) = \text{softmax}(W_U N(h^{(\ell)})). \quad (2)$$

Definition 2.2 applies the trained unembedding to an intermediate residual state, which is the logit lens of [7]. By construction $p_L = p_M$; no claim is made that p_ℓ is calibrated for $\ell < L$, and indeed Remark 3.26 is that it is not.

Definition 2.3 (Residual tail energy). The *tail energy* of M at layer ℓ on prefix x is

$$T_\ell(x) = \left\| \sum_{k=\ell+1}^L F_k(h^{(k-1)}) \right\|_2 = \|h^{(L)} - h^{(\ell)}\|_2, \quad (3)$$

the norm of the residual contribution the model has yet to make after layer ℓ . The second equality is immediate by telescoping (1).

2.2 A cost model for autoregressive decoding

Decoding cost is dominated by different resources in different regimes, and conflating them is the most common error in speculative-decoding accounting. Evaluating a block at m positions requires streaming its parameters from memory once, regardless of m , and performing work proportional to m . We therefore make the regime an explicit parameter.

Definition 2.4 (Interpolated cost model). Fix $w \in [0, 1]$. The cost of evaluating Δ consecutive blocks of M at m positions, in units of one full forward pass at a single position, is

$$C_w(\Delta, m) = \frac{\Delta}{L} (w + (1 - w) m). \quad (4)$$

The parameter w is the *memory-bound fraction*: $w = 1$ is the strictly memory-bound regime (cost is weight streaming alone, independent of m), $w = 0$ the strictly compute-bound regime (cost is proportional to m). Batch-size-one autoregressive decoding operates at w close to 1; prefill and large-batch serving at w close to 0. Note $C_w(\Delta, 1) = \Delta/L$ for every w .

Remark 2.5. Equation (4) is deliberately the simplest interpolation with the right two endpoints. It ignores attention’s quadratic term and KV-cache traffic, both of which grow with context and neither of which affects the comparisons in Section 3, since they are incurred identically by every scheme considered there.

2.3 Speculative sampling

Definition 2.6 (Speculative round). Let p be a target distribution and q a *draft* distribution on $\mathcal{V}$. A *speculative step* draws $y \sim q$, accepts it with probability $\min\{1, p(y)/q(y)\}$, and on rejection draws instead from the residual

$$r(\cdot) = \frac{(p(\cdot) - q(\cdot))_+}{\|(p - q)_+\|_1}, \quad (5)$$

where $(z)_+ = \max\{z, 0\}$. A *round with lookahead* γ performs γ such steps against drafts for successive positions, stopping at the first rejection, and emits every accepted token plus one token drawn from the target (or from (5) on rejection).

Definition 2.6 leaves r undefined when $\|(p - q)_+\|_1 = 0$, which occurs exactly when $q = p$; in that case acceptance is certain and r is never sampled, so we adopt the convention $r = p$ there.

2.4 Chain-of-thought and latent recurrence

The two computational media compared in Section 4 are separated here so that the comparison in that section is between definitions rather than between implementations.

Definition 2.7 (Chain-of-thought trajectory). Fix a model M and a prompt $x \in \mathcal{V}^*$. An *n -step chain-of-thought trajectory* is the sequence of emitted tokens $y_{1:n} \in \mathcal{V}^n$ produced by $y_i \sim p_M(\cdot \mid x y_{1:i-1})$, together with the induced internal states. The *configuration* after step i is the tuple of all activations and cache entries of M on the input $x y_{1:i}$.

Definition 2.8 (Latent-recurrent computation). An n -step *latent-recurrent computation* of width d is the sequence $s_1, \dots, s_n$ given by $s_0 = \iota(x)$ and $s_i = G(s_{i-1})$ for a map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$, with the output read off by a decoder $\pi(s_n)$. Its *configuration* after step i is s_i .

Definition 2.9 (Reachable configuration set). For a fixed prompt x and a computational medium, $\text{Reach}_n(x)$ denotes the set of configurations reachable after exactly n steps, as the trajectory ranges over all its allowed realisations.

Definition 2.9 is deliberately a set, not a distribution. Every separation we prove in Section 4 is a cardinality statement about Reach_n , and none requires a notion of entropy; we avoid Shannon information entirely there, because the quantity that binds is how many configurations a medium can *distinguish*, not how many bits it carries on average.

2.5 Recurrent form

Definition 2.10 (Recurrent form and state dimension). A causal sequence model is in *recurrent form* if there are maps f, g and states S_t with

$$S_t = f(S_{t-1}, x_t), \quad o_t = g(S_t, x_t). \quad (6)$$

Its *state dimension* $\dim(S_t)$ is the number of scalars in S_t . The form is *bounded* if $\sup_t \dim(S_t) < \infty$ and *growing* otherwise.

The distinction in Definition 2.10 between bounded and growing is the only one that carries content; Remark 5.1 makes precise the sense in which the existence of *some* recurrent form is trivial.

3 Depth: the model as its own draft

The complaint addressed in this section is that speculative decoding [1, 2] requires a second, separately trained network. We show that correctness never required one (Theorem 3.1), that the acceptance rate of an early-exit draft is controlled by a measurable property of the residual stream (Theorem 3.7), and that under the cost model of Definition 2.4 self-drafting strictly dominates any external draft of equal relative cost, in *every* regime w (Theorem 3.20).

3.1 Correctness does not constrain the draft

Theorem 3.1 (Exactness for an arbitrary draft). *Let p, q be any distributions on a common countable alphabet $\mathcal{V}$, with the convention $r = p$ when $q = p$. Then the output of a speculative step (Definition 2.6) is distributed exactly as p , and the acceptance probability is*

$$\mathbb{P}[\text{accept}] = \sum_x \min\{p(x), q(x)\} = 1 - \text{TV}(p, q). \quad (7)$$

Proof. Fix $x \in \mathcal{V}$. The token x is emitted either by being drawn from q and accepted, or by being drawn from the residual (5) after a rejection. The first event has probability $q(x) \min\{1, p(x)/q(x)\} = \min\{p(x), q(x)\}$, with the convention that the term is 0 when $q(x) = 0$. Summing over x ,

$$\mathbb{P}[\text{accept}] = \sum_x \min\{p(x), q(x)\} = \sum_x \left(p(x) - (p(x) - q(x))_+ \right) = 1 - \|(p - q)_+\|_1, \quad (8)$$

so $\mathbb{P}[\text{reject}] = \|(p - q)_+\|_1$, and the second event has probability $\|(p - q)_+\|_1 \cdot (p(x) - q(x))_+ / \|(p - q)_+\|_1 = (p(x) - q(x))_+$. Hence

$$\mathbb{P}[\text{emit } x] = \min\{p(x), q(x)\} + (p(x) - q(x))_+ = p(x), \quad (9)$$

using $\min\{a, b\} + (a - b)_+ = a$. Finally $\sum_x \min\{p, q\} = 1 - \frac{1}{2}\|p - q\|_1$ follows from $\min\{a, b\} = \frac{1}{2}(a + b - |a - b|)$ and $\sum_x p = \sum_x q = 1$. No support hypothesis is required: when $q(x) = 0 < p(x)$ the token is unreachable by acceptance, contributes 0 to it, and is supplied by the residual branch with exactly the right mass $(p(x) - q(x))_+ = p(x)$. $\square$

Remark 3.2 (The draft is a free parameter). Theorem 3.1 places *no* condition on q whatsoever: q may be a smaller model, a layer- ℓ readout of M itself (Definition 2.2), an n -gram table, or noise. The output distribution is p regardless, and only the *rate* (7) changes. Consequently no argument from correctness supports maintaining a separate draft network; the entire design question is the ratio of acceptance to cost, which is what Section 3.3 formalises.

3.2 Acceptance is controlled by residual tail energy

Lemma 3.3 (Softmax is TV-Lipschitz in the spread seminorm). *For $z \in \mathbb{R}^V$ write $\text{spr}(z) = \max_j z_j - \min_j z_j$. Then for all $u, v \in \mathbb{R}^V$, with $\sigma = \text{softmax}$,*

$$\text{TV}(\sigma(u), \sigma(v)) \leq 1 - e^{-\text{spr}(u-v)} \leq 1 - e^{-2\|u-v\|_\infty}. \quad (10)$$

Proof. First suppose $\varepsilon = \|u-v\|_\infty$. For every i , $\sigma(u)_i / \sigma(v)_i = e^{u_i - v_i} \sum_j e^{v_j} / \sum_j e^{u_j}$, and since $|u_j - v_j| \leq \varepsilon$ for all j we get $e^{-\varepsilon} \sum_j e^{v_j} \leq \sum_j e^{u_j} \leq e^\varepsilon \sum_j e^{v_j}$ and $e^{u_i - v_i} \in [e^{-\varepsilon}, e^\varepsilon]$. Multiplying, $\sigma(u)_i \geq e^{-2\varepsilon} \sigma(v)_i$, so by $\text{TV}(p, q) = \sum_i (q_i - p_i)_+$,

$$\text{TV}(\sigma(u), \sigma(v)) = \sum_i (\sigma(v)_i - \sigma(u)_i)_+ \leq \sum_i \sigma(v)_i (1 - e^{-2\varepsilon}) = 1 - e^{-2\varepsilon}. \quad (11)$$

Now sharpen using shift invariance. Softmax satisfies $\sigma(u) = \sigma(u + c\mathbf{1})$ for every $c \in \mathbb{R}$, so with $z = u - v$, $s = \text{spr}(z)$ and $c = -\frac{1}{2}(\max_j z_j + \min_j z_j)$ we have $\sigma(u) = \sigma(v + (z + c\mathbf{1}))$ and $\|z + c\mathbf{1}\|_\infty = s/2$. Applying (11) with $\varepsilon = s/2$ gives $\text{TV} \leq 1 - e^{-s}$. Finally $s \leq 2\|z\|_\infty$. $\square$

Remark 3.4. The sup-norm form (11) is the one usually written down, but $\|u - v\|_\infty$ is not shift invariant while $\text{TV}(\sigma(u), \sigma(v))$ is, so it is slack by up to a factor of two — an artificial factor that would otherwise propagate into every exponent below.

Lemma 3.5 (Sharp bound for normalisation). *For all non-zero $a, b \in \mathbb{R}^d$,*

$$\left\| \frac{a}{\|a\|} - \frac{b}{\|b\|} \right\| \leq \frac{2\|a - b\|}{\|a\| + \|b\|}. \quad (12)$$

Consequently, for RMSNorm $N(h) = \sqrt{d} \text{Diag}(g) h / \|h\|$,

$$\|N(h) - N(h')\| \leq \frac{2\sqrt{d}\|g\|_\infty}{\|h\| + \|h'\|} \|h - h'\|. \quad (13)$$

Proof. Put $s = \|a\|$, $t = \|b\|$ and $c = \langle a, b \rangle / (st) \in [-1, 1]$. The squared left side is $2 - 2c$ and the squared right side is $4(s^2 + t^2 - 2stc) / (s + t)^2$. Their difference, after multiplying by $(s + t)^2 > 0$, is

$$\begin{aligned} 4(s^2 + t^2 - 2stc) - (2 - 2c)(s + t)^2 &= 2s^2 + 2t^2 - 4st - 4stc + 2cs^2 + 2ct^2 \\ &= 2(s - t)^2 + 2c(s - t)^2 = 2(1 + c)(s - t)^2 \geq 0, \end{aligned} \quad (14)$$

since $c \geq -1$. The RMSNorm form follows from $N(h) - N(h') = \sqrt{d} \text{Diag}(g)(h/\|h\| - h'/\|h'\|)$ and $\|\text{Diag}(g)z\| \leq \|g\|_\infty \|z\|$. $\square$

Remark 3.6. Lemma 3.5 is an exact inequality between the two endpoints and requires no Lipschitz hypothesis. This matters: a bound of the form “ N is κ -Lipschitz on a region containing $h^{(\ell)}$ and $h^{(L)}$ ” is vacuous as usually stated, since a two-point set admits every constant, and repairing it requires a convex region whose distance to the origin — not the endpoint norms — sets the constant.

Theorem 3.7 (Acceptance lower bound). *Let $\alpha_\ell = 1 - \text{TV}(p_\ell, p_L)$ be the acceptance rate of the layer- ℓ readout used as a draft for M , let T_ℓ be the tail energy (Definition 2.3), let N be RMSNorm with gain g , and define the unembedding diameter*

$$D_U = \max_{i,j} \|(W_U)_{i,:} - (W_U)_{j,:}\|_2 \leq 2\|W_U\|_{2,\infty}. \quad (15)$$

Then

$$\alpha_\ell \geq \exp\left(-\frac{2\sqrt{d}\|g\|_\infty D_U T_\ell}{\|h^{(\ell)}\|_2 + \|h^{(L)}\|_2}\right). \quad (16)$$

Proof. Write $D = N(h^{(\ell)}) - N(h^{(L)})$ and $w_i = (W_U)_{i,:}$, so that the logit difference has coordinates $\langle w_i, D \rangle$. By Lemma 3.3 it suffices to bound the spread:

$$\text{spr}(W_U D) = \max_{i,j} \langle w_i - w_j, D \rangle \leq \max_{i,j} \|w_i - w_j\| \|D\| = D_U \|D\|, \quad (17)$$

by Cauchy–Schwarz. By Lemma 3.5 applied to $h = h^{(\ell)}$, $h' = h^{(L)}$ and Definition 2.3, $\|D\| \leq 2\sqrt{d}\|g\|_\infty T_\ell / (\|h^{(\ell)}\| + \|h^{(L)}\|)$. Combining and using $\alpha_\ell \geq e^{-\text{spr}}$ gives the claim. $\square$

Remark 3.8 (Strictly stronger than the sup-norm form). Bound (16) improves the naive estimate on both factors: D_U replaces $2\|W_U\|_{2,\infty}$ because only *differences* of unembedding rows affect a softmax, and $\|h^{(\ell)}\| + \|h^{(L)}\|$ replaces a single endpoint norm. It is also scale-free, as it must be, since N discards the scale of its argument.

Remark 3.9 (What the bound is for). Bound (16) is monotone in the right quantity — acceptance is governed by how much work the model has *left* to do — but we expect it to be numerically weak. Two inequalities give information away, and they can be measured separately: Cauchy–Schwarz with the diameter D_U assumes the worst pair of unembedding rows aligns with the tail direction, and Lemma 3.3 is tight only when the logit gap is spread uniformly across the vocabulary. We had expected the first to dominate; Section 6 reports that this expectation was wrong.

Proposition 3.10 (Unconditional subspace refinement). *Let P be any orthogonal projector and $Q = I - P$. Then*

$$\alpha_\ell \geq \exp(-M_P(D)), \quad M_P(D) = \max_{i,j} \left(\|P(w_i - w_j)\| \|PD\| + \|Q(w_i - w_j)\| \|QD\| \right), \quad (18)$$

and $M_P(D) \leq D_U \|D\|$ always, so this is never worse than Theorem 3.7.

Proof. Orthogonality gives $\langle w_i - w_j, D \rangle = \langle P(w_i - w_j), PD \rangle + \langle Q(w_i - w_j), QD \rangle$, and Cauchy–Schwarz on each term gives the bound on the spread. For the comparison, apply Cauchy–Schwarz in $\mathbb{R}^2$ to the vectors $(\|P(w_i - w_j)\|, \|Q(w_i - w_j)\|)$ and $(\|PD\|, \|QD\|)$, whose norms are $\|w_i - w_j\|$ and $\|D\|$ respectively. $\square$

Remark 3.11 (Why the one-term version fails). It is tempting to drop the Q term, claiming that if the tail concentrates in $\text{im } P$ then $\text{spr} \leq A_P \|D\|$ with $A_P = \max_{i,j} \|P(w_i - w_j)\|$. This is *false* in general. Writing B_P for the corresponding Q -norm and $c = \|PD\|^2 / \|D\|^2$ for the captured fraction, the one-term bound holds iff $B_P \sqrt{1-c} \leq A_P(1-\sqrt{c})$; for small leakage $\eta = \sqrt{1-c}$ the right side shrinks like $\eta^2/2$ while the left shrinks like η , so the condition is roughly $B_P/A_P \lesssim \eta/2$. Capturing 99% of the tail energy therefore still requires the unembedding rows themselves to be 95% contained in the same subspace — a joint condition on W_U and the tail, not a statement about the tail alone. Proposition 3.10 avoids the issue by keeping both terms, and is valid for a P fitted to the very data it is evaluated on, being a deterministic per-example inequality. Note also that P must be built from the *post-normalisation* difference D , not from $h^{(L)} - h^{(\ell)}$, since RMSNorm’s rescaling and gain change the relevant subspace.

3.3 Cost accounting, and when self-drafting dominates

We first record an exact expression for round yield that makes no independence assumption, since acceptance events at successive positions are strongly correlated in practice.

Proposition 3.12 (Exact expected yield). *Consider a round with lookahead γ (Definition 2.6) and let A_j be the event that the first j drafted tokens are all accepted, $j = 1, \dots, \gamma$. Then the number Y of tokens emitted in the round satisfies*

$$Y = 1 + \sum_{j=1}^{\gamma} \mathbf{1}[A_j], \quad \text{hence} \quad \mathbb{E}[Y] = 1 + \sum_{j=1}^{\gamma} \mathbb{P}[A_j]. \quad (19)$$

Proof. Acceptance halts at the first rejection, so the number of accepted tokens is $K = \max\{j : A_j \text{ holds}\}$ (with $K = 0$ if A_1 fails), and A_j holds if and only if $j \leq K$. Therefore $\sum_{j=1}^{\gamma} \mathbf{1}[A_j] = K$. The round emits the K accepted tokens plus exactly one further token — drawn from the residual (5) if $K < \gamma$, and from the target at the final position if $K = \gamma$ — so $Y = K + 1$. Linearity of expectation gives the second identity; no independence is used. $\square$

Corollary 3.13 (Geometric special case). *If acceptance events are i.i.d. with rate α , then $\mathbb{P}[A_j] = \alpha^j$ and $\mathbb{E}[Y] = (1 - \alpha^{\gamma+1})/(1 - \alpha)$.*

Proof. Substitute into (19): $\mathbb{E}[Y] = 1 + \sum_{j=1}^{\gamma} \alpha^j = 1 + \frac{\alpha - \alpha^{\gamma+1}}{1 - \alpha} = \frac{1 - \alpha^{\gamma+1}}{1 - \alpha}$. $\square$

Remark 3.14. Correlated acceptance makes $\mathbb{P}[A_j] \geq \prod_{i \leq j} \alpha_i$ typical — easy contexts accept runs, hard contexts reject early — so Corollary 3.13 evaluated at the mean acceptance rate generally *under-estimates* yield. Reported speedups computed from Corollary 3.13 are therefore conservative, which is the safe direction for the comparison below, since both schemes are scored the same way.

Definition 3.15 (Head fraction). Let $u \in [0, 1)$ be the fraction of a full single-position forward pass attributable to the unembedding W_U , so that the L blocks jointly account for $1 - u$. Every draft step must evaluate W_U , since a draft is useless unless a token can be sampled from it.

Proposition 3.16 (Round cost). *Let $\rho = \ell/L$ and $D_k = w + (1 - w)k$. Drafting evaluates blocks $1..\ell$ at positions $t, \dots, t + \gamma - 1$. Verification requires full depth at all of $t, \dots, t + \gamma$; the final position $t + \gamma$, which supplies the bonus token, is never touched by drafting. Two verification strategies exist:*

$$C_{\text{self}}^{\text{nr}} = \gamma[(1 - u)\rho + u] + D_{\gamma+1}, \quad (20)$$

$$C_{\text{self}}^{\text{re}} = \gamma[(1 - u)\rho + u] + (1 - u)(1 - \rho)D_{\gamma} + (1 - u) + uD_{\gamma+1}, \quad (21)$$

$$C_{\text{ext}} = \gamma c_d + D_{\gamma+1}, \quad (22)$$

and $C_{\text{self}} = \min\{C_{\text{self}}^{\text{nr}}, C_{\text{self}}^{\text{re}}\}$.

Proof. Each of the γ draft steps evaluates blocks $1..\ell$ and the head at a single position, costing $(1 - u)\rho + u$ independently of w . For (20), verification is one batched full-depth pass over the $\gamma + 1$ positions, costing $D_{\gamma+1}$. For (21), cached keys and values let the γ drafted positions resume at block $\ell + 1$, costing $(1 - u)(1 - \rho)D_{\gamma}$, but position $t + \gamma$ has nothing cached and needs a full-depth pass of its own, costing $(1 - u)$; the head is evaluated at all $\gamma + 1$ positions. The external draft shares no blocks with M . $\square$

Remark 3.17 (Sanity check, and an error it catches). At $\rho = 1$ the draft is the target, so $\alpha = 1$ and the scheme degenerates to ordinary autoregressive decoding, forcing $S = 1$. Both (20) and (21) give $C = \gamma + 1$ there at $w = 1$, matching the yield $\gamma + 1$. A cost model that omits the bonus position instead returns $S = (\gamma + 1)/(\gamma + u) > 1$, manufacturing speedup from nothing. An earlier version of this paper contained exactly that error, and it was caught only by running the experiment.

Proposition 3.18 (Cache reuse is regime-dependent, and useless when memory-bound). *At $w = 1$, $C_{\text{self}}^{\text{re}} \geq C_{\text{self}}^{\text{nr}}$ for every $\rho \in [0, 1]$, with equality only at $\rho = 1$. At $w = 0$ the inequality reverses for every $\rho > 0$.*

Proof. At $w = 1$ we have $D_k = 1$ for all k , so $C^{\text{re}} - C^{\text{nr}} = (1-u)(1-\rho) + (1-u) + u - 1 = (1-u)(1-\rho) \geq 0$. At $w = 0$, $D_k = k$, and $C^{\text{re}} - C^{\text{nr}} = (1-u)[(1-\rho)\gamma + 1 - (\gamma + 1)] = -(1-u)\rho\gamma < 0$. $\square$

Remark 3.19 (Retraction). An earlier version argued that cache reuse is the structural advantage of self-drafting, since skipping blocks $1..\ell$ also skips streaming their parameters. The reasoning is incomplete and the conclusion is wrong: reuse splits verification into two passes at *different depths*, and in the memory-bound regime a pass costs a full weight stream however many positions it covers, so the lower blocks are paid for twice. At $u = 0$, $\rho = 0.25$, $\gamma = 2$ reuse costs 2.25 against 1.50 without it. The genuine advantage of self-drafting when memory-bound is simply that there is no second model to stream; reuse belongs to the compute-bound case.

Theorem 3.20 (Self-drafting dominates). *Fix $\gamma \geq 1$, $w \in [0, 1]$, $\rho \in (0, 1)$. If an external draft model has relative cost $c_d \geq u + (1-u)\rho$ and acceptance $\alpha_d \leq \alpha_\ell$, then early-exit self-drafting at layer ℓ attains at least as great a speedup, and strictly greater when $c_d > u + (1-u)\rho$.*

Proof. Comparing (22) with (20), $C_{\text{ext}} - C_{\text{self}}^{\text{nr}} = \gamma[c_d - u - (1-u)\rho] \geq 0$, with equality exactly when $c_d = u + (1-u)\rho$, and $C_{\text{self}} \leq C_{\text{self}}^{\text{nr}}$. Yield is non-decreasing in the acceptance events by Proposition 3.12, so $\mathbb{E}[Y \mid \alpha_\ell] \geq \mathbb{E}[Y \mid \alpha_d] > 0$. $\square$

Remark 3.21. The strictness claimed in an earlier version came from the reuse term retracted in Remark 3.19. What survives is weak dominance. Not modelled, and further favouring the self-draft: an external draft occupies additional resident memory and requires vocabulary alignment between the two models.

Corollary 3.22 (The head is a hard ceiling). *Even with perfect acceptance and $\rho \rightarrow 0$, the speedup of self-drafting satisfies*

$$S \leq \frac{\gamma + 1}{\gamma u + D} \xrightarrow{w \rightarrow 1} \frac{\gamma + 1}{\gamma u + 1}. \quad (23)$$

For Qwen3-0.6B, whose tied unembedding is 151936×1024 of roughly 5.96×10^8 parameters, $u \approx 0.26$, giving $S \leq 2.45$ at $\gamma = 4$.

Remark 3.23 (Why omitting the head is not a rounding error). Writing $C_{\text{self}} = \gamma\rho + 1$, as is common, assumes a draft step streams only the first ℓ blocks. It must also stream W_U . At $\gamma = 4$ the omission inflates the reported speedup by a factor of 1.07 at $\rho = 0.75$, 1.17 at $\rho = 0.5$ and 1.39 at $\rho = 0.25$ — largest exactly where early exit is most attractive. Since our pre-registered kill criterion for this thesis is $\max S \leq 1$, an error of this size and sign can invert the conclusion, and small models with large vocabularies are the worst case rather than a corner case.

Corollary 3.24 (Shortlisted draft head). *Let $S \subseteq \mathcal{V}$ be chosen independently of the current logits, and let the draft be $q_S(x) = p_\ell(x)/p_\ell(S)$ for $x \in S$ and 0 otherwise. Then decoding remains exact, the head cost falls from u to $u|S|/V$, and*

$$\alpha_S \geq \alpha_\ell - p_L(\mathcal{V} \setminus S). \quad (24)$$

Proof. Exactness is Theorem 3.1, which requires no support condition — this is where its full generality is used. For the rate, $q_S \geq p_\ell$ pointwise on S , so $\alpha_S = \sum_{x \in S} \min\{p_L, q_S\} \geq \sum_{x \in S} \min\{p_L, p_\ell\} = \alpha_\ell - \sum_{x \notin S} \min\{p_L, p_\ell\} \geq \alpha_\ell - p_L(S^c)$. The cost claim is immediate, as only the rows of W_U indexed by S are needed. $\square$

Remark 3.25. S must be fixed in advance — a static frequency shortlist, or one conditioned on the previous token — because selecting a top- k set requires the full logit vector and so would not save the streaming of W_U . Corollary 3.24 is the natural route around Corollary 3.22, and it trades a bounded, *measurable* acceptance loss for the head cost.

Remark 3.26 (The real obstruction is the training objective). Theorem 3.20 is conditional on $\alpha_\ell \geq \alpha_d$, and for an untrained early exit this is exactly what fails: W_U was fit against $N(h^{(L)})$ alone, so $N(h^{(\ell)})$ is off-distribution for it and α_ℓ is small at useful ρ . This is a statement about the training objective rather than the architecture, and is what an early-exit auxiliary loss with layer dropout addresses [3, 4]. The complaint of this section is therefore best stated not as “speculative decoding is a hack” but as: *models are trained so that their intermediate layers are not decodable, and a second network is then introduced to supply what the first was never asked to expose*.

4 Time: serial computation and the width of its carrier

The complaint addressed here is that chain-of-thought is a workaround for a model that never learned to reason latently. We show the strong form is false under standard assumptions (Section 4.1); that the comparison between media is a genuine trade-off with a computable crossover, but in the *opposite* direction to the one usually asserted (Section 4.2); and that an unconditional separation is available only in a restricted read model, with a barrier explaining why (Section 4.3).

4.1 Serial computation is load-bearing

Theorem 4.1 (Depth is not a substitute for steps). *Let $\mathcal{T}$ be the class of languages decided by fixed-depth, log-precision transformer decoders whose feedforward maps are computable in space linear in their input, and let $\mathcal{T}_{\text{CoT}}$ be the class decided by the same decoders granted $\text{poly}(n)$ chain-of-thought steps. Then $\mathcal{T} \subseteq \text{uniform TC}^0$ [8] and $\mathcal{T}_{\text{CoT}} = \text{P}$ [9]. Consequently, if $\text{TC}^0 \neq \text{P}$, then $\mathcal{T} \subsetneq \mathcal{T}_{\text{CoT}}$.*

Remark 4.2 (Three hypotheses that are usually dropped). First, the uniformity in [8] is a condition on the *model* — the feedforward maps must be space-constructible — not a free adjective on TC^0 ; without it the containment is in non-uniform TC^0 and the separation required becomes $\text{TC}^0/\text{poly} \neq \text{P}$. Second, these are statements about *classes of languages*, not about a particular network: a single trained model decides one language, so “this model cannot do it without CoT” does not follow from Theorem 4.1 for any individual model, only the non-existence of *some* fixed-depth decoder for the hard language. Third, the “= P” direction additionally requires the generalized pre-norm architecture of [9]. With these in place the conclusion still stands, and it is fatal to the strong form of the complaint: serial steps buy computational power that no amount of latent skill at fixed depth recovers.

4.2 Scratchpad capacity: a trade-off, not a dominance

We now compare media. The natural intuition — a token carries $\log_2 |\mathcal{V}| \approx 17$ bits, a hidden state carries thousands, therefore latent recurrence transports far more per step — is correct about *per-step width* and wrong about *total state*, because the two media differ in whether that width accumulates.

Definition 4.3 (Step medium). A *step medium* of width b is a mechanism by which one serial step transmits information to subsequent steps. It is *accumulating* if the contents written at every step remain readable at all later steps, and *non-accumulating* if each step’s write replaces the previous contents.

Chain-of-thought is accumulating with $b = \log_2 V$: the emitted tokens persist in the context and every later step attends over all of them. A looped or depth-recurrent model carrying a single state $s_i \in \mathbb{R}^d$ is non-accumulating with $b = d b_{\text{eff}}$, where b_{eff} is the effective bits per coordinate. A latent model that *appends* each continuous thought to the sequence, as in [12], is accumulating with $b = d b_{\text{eff}}$.

Theorem 4.4 (Scratchpad capacity). *After n steps, the set of distinguishable carrier contents has cardinality at most 2^{nb} for an accumulating medium of width b , and at most 2^b for a non-accumulating one.*

Proof. An accumulating medium’s contents after n steps are the tuple of the n writes, each drawn from a set of size at most 2^b . A non-accumulating medium’s contents are the last write alone, from a set of size at most 2^b ; iterating a map cannot increase cardinality, since $|G(A)| \leq |A|$ for any G and any A . $\square$

Corollary 4.5 (Crossover). *Chain-of-thought carries strictly more state than a fixed-size latent loop once*

$$n > n^* = \frac{d b_{\text{eff}}}{\log_2 V}. \quad (25)$$

For $d = 1024$, $b_{\text{eff}} = 4$ and $V = 151936$ this is $n^* \approx 238$ steps. Against an accumulating latent medium of the same width, by contrast, chain-of-thought is dominated at every n .

Remark 4.6 (This reverses the usual claim, and removes a contradiction). The informal argument — 17 bits against thousands, therefore latent recurrence wins — silently compares a per-step width with a per-step width while the quantity that matters for a long derivation is total accessible state. A fixed-size latent loop *overwrites*; a chain of tokens *appends*. Past a few hundred steps, which is exactly the regime long chains of thought occupy, the token stream is the larger memory. This is the same phenomenon as Theorem 5.12 viewed from the other side: bounded state is a liability for long-horizon work, and it does not stop being one because the state is continuous. The defensible form of the complaint is therefore narrower and sharper than the original: *the discretisation is a real cost per step, and it is worth paying only where the medium accumulates*. Latent recurrence should append, not overwrite — and a model that appends continuous thoughts dominates chain-of-thought on this measure at every horizon.

4.3 Simulation, and where a separation can live

Theorem 4.7 (Latent recurrence simulates chain-of-thought). *Fix a model M with L -Lipschitz blocks, a finite prompt set, and a decoding temperature $\tau > 0$, and suppose the logit margin at every step of the trajectory is at least $\delta > 0$. Then there is an n -step latent-recurrent computation (Definition 2.8) whose induced token sequence equals the n -step chain-of-thought trajectory of M exactly.*

Proof sketch. Take G to be: run M ’s stack on the current state, apply $W_U N(\cdot)$, form $\text{softmax}(\cdot/\tau)$, and write back the corresponding convex combination of embeddings. Under a uniform margin δ , taking τ small enough makes this softmax within any desired ε of the one-hot at the argmax, and finiteness of the prompt set makes the required τ uniform; the Lipschitz hypothesis converts the ε on the written state into a bound on the downstream deviation, which stays below $\delta/2$ and hence preserves every subsequent argmax. Each G is a composition of M ’s own maps, so the computation lies in the latent-recurrent class. $\square$

Remark 4.8 (A precision tension). Theorem 4.7 requires $\Omega(n \log L)$ bits of precision to keep the margin through n steps, whereas Theorem 4.1 is stated for log-precision models. The two results are therefore about different machines, and no single model satisfies both hypotheses for large n . We state this rather than paper over it.

Remark 4.9 (Strictness does not follow). Theorem 4.7 gives an inclusion. The converse separation does *not* follow from Theorem 4.4, which after Corollary 4.5 runs the other way for large n ; and a naive counting separation fails outright, because intermediate states of a deterministic task are functions of the prompt. Concretely, a prompt containing two permutations of S_{30} induces $30!$ distinct “intermediate configurations”, which a counting argument would say requires seven CoT tokens to distinguish, while a model in fact distinguishes them at $n = 0$ by attending to the prompt.

The escape in Remark 4.9 is re-reading the prompt, and it must be closed by hypothesis rather than by choice of task.

Theorem 4.10 (Separation under a streaming read schedule). *Let $\text{PROD}_{N,T}$ be the problem of composing T given elements of S_N . Consider a step machine that reads one input symbol per step, may perform arbitrary computation within a step, and carries information between steps only through an interface of width b bits. Such a machine solves $\text{PROD}_{N,T}$ for all T if and only if $b \geq \log_2(N!)$.*

Proof. Sufficiency: carry the running product, which is an element of S_N and so fits in $\lceil \log_2 N! \rceil$ bits; each step composes it with the incoming generator. Necessity: the induced map from input prefixes to interface contents must be constant on Myhill–Nerode classes of the language $\{(g_{1:T}, \pi) : g_T \cdots g_1 = \pi\}$. Two prefixes with distinct running products are inequivalent, since a suitable suffix distinguishes them, so the machine needs at least $|S_N| = N!$ interface states. $\square$

Corollary 4.11. *With $V = 151936$ and $db_{\text{eff}} = 4096$, a one-token-per-step CoT interface fails from $N = 9$ onwards, since $\log_2 8! = 15.30 < \log_2 V = 17.21 < 18.47 = \log_2 9!$, while the latent interface suffices to $N = 536$.*

Proposition 4.12 (The streaming hypothesis is doing real work). *With full access to the prompt and unbounded computation within a step, a single-step machine solves $\text{PROD}_{N,T}$ for every T .*

Proof. The entire input is available in one step; compute the product. $\square$

Remark 4.13 (A barrier). Proposition 4.12 means any separation in the full-attention setting must be a circuit-complexity statement, not a counting one — and such statements are currently out of reach. By Barrington’s theorem, word problems over S_5 are NC^1 -complete, so a theorem of the form “ $\text{PROD}_{N,T}$ requires at least two chain-of-thought steps for a fixed-depth transformer” would imply $\text{TC}^0 \neq \text{NC}^1$. The bandwidth separation of Theorem 4.10 is therefore the strongest unconditional result available, and it is conditional on a read schedule that real architectures do not obey. Our experimental protocol (Section 8) must enforce that schedule rather than assume it.

4.4 Discretisation as error correction

Open Problem 4.14 (Bandwidth against drift). Fix a latent update G and per-step perturbation of variance η^2 , and let $\mathcal{C}$ be a codebook of rate R bits per step onto which the state is projected every k steps. Quantify the largest n for which the trajectory remains recoverable, as a function of (η, R, k) .

We can settle the simplest instance exactly, and it does not contain the trade-off the problem presupposes.

Proposition 4.15 (Projection interval with a rate-sufficient codebook). *Store a bit as the sign of a coordinate initialised at ± 1 , add $\mathcal{N}(0, \sigma^2)$ noise at each of T steps, and every k steps optionally project to the nearest of $\{-1, +1\}$. With $p_k = \Phi(-1/(\sigma\sqrt{k}))$,*

$$\mathbb{P}[\text{correct} \mid \text{project every } k] = \frac{1}{2} \left(1 + (1 - 2p_k)^{T/k} \right), \quad \mathbb{P}[\text{correct} \mid \text{never}] = 1 - \Phi(-1/(\sigma\sqrt{T})). \quad (26)$$

Proof. After a projection the state is exactly ± 1 ; over the next k steps it accumulates $\mathcal{N}(0, k\sigma^2)$ and is projected to the wrong sign iff that sum crosses ∓ 1 , with probability p_k . The same holds from the wrong sign, so the projected state is a two-state Markov chain with symmetric flip probability p_k per interval, whose T/k -step transition gives the first formula. Without projection the final value is $1 + \mathcal{N}(0, T\sigma^2)$. $\square$

Two consequences. First, projection does *not* lock errors in: a flipped bit flips back with the same probability, which is why the chain is symmetric. The lock-in penalty conjectured in an earlier draft of this paper does not exist in this model. Second, evaluating (26) over $\sigma \in [0.05, 0.4]$, $k \in \{1, \dots, 32\}$, $T = 64$ finds $k = 1$ optimal at every noise level, confirmed by Monte Carlo at 2×10^4 trials (Section 6); in the small- σ regime this follows from p_k/k being increasing in k , so that $(T/k) \ln(1 - 2p_k)$ is decreasing. With enough rate, discretise as often as possible.

The trade-off appears only when the codebook’s resolution is coarser than the task requires, and even then it is binary rather than interior: with an analog value to be recovered to precision ε and a grid of L levels with half-spacing $\Delta/2 > \varepsilon$, Monte Carlo finds “never project” optimal while $\sigma\sqrt{T}$ is below the rounding error and “project every step” optimal once drift exceeds it, with no clear interior- k winner in any cell. The design question, in this model, is therefore not how often to discretise but whether the codebook rate clears

the task’s precision requirement — above it, always; below it, only once drift dominates. Whether a learned codebook inside a trained network behaves the same way is the part of Open Problem 4.14 that remains open. Our attempt to test it (Section 6) was defeated by optimisation rather than answered: with a 16-bit learned codebook carrying 8 bits through 64 identity-plus-noise steps, accuracy at $\sigma = 0$ — where projection is information-theoretically free — was monotone in the *number* of projections ($64 \rightarrow 0.48$, $16 \rightarrow 0.50$, $4 \rightarrow 0.75$, $0 \rightarrow 1.00$). That is the signature of straight-through gradients failing through stacked quantisations, not of error correction, and it says nothing about the question either way. A codebook that is a stable fixed point of its own projection is what the learned setting needs and what plain SGD did not find; a curriculum that introduces projection after the unprojected task is learned is the obvious next attempt.

5 State: recurrent form, the HMM reading, and its limits

The complaint addressed here is that decoder-only transformers are “just gated RNNs”, and that constraining such an RNN should yield HMM-like decoding. We find the first half true but nearly empty as usually stated (Section 5.1), the HMM identity true but *not* a statement about linear attention (Section 5.3), and we prove that the missing bridge cannot be built (Theorem 5.8). We close with the capacity bound that says what unbounded state buys (Theorem 5.12).

5.1 The recurrent form is real, and almost vacuous

Remark 5.1 (The trivial recurrence). Every causal sequence model admits a growing-state recurrent form: take $S_t = x_{1:t}$ and $o_t = M(S_t)$. So the assertion “a decoder-only transformer is a recurrent network with growing state” carries exactly the content of “a decoder-only transformer is causal”. Worse for the usual framing, the trivial state is *smaller* than the KV cache — $t \log_2 V$ bits against $\Theta(t d L)$ scalars — so a transformer does not *compress* its history into its state; it expands it.

The content is therefore not existence but cost, and the theorem should be stated that way.

Theorem 5.2 (KV cache as an incrementally maintainable sufficient statistic). *Let M be a decoder-only transformer with L layers, width d , and H_{kv} key-value heads of dimension d_h . Put $S_t = (K_{1:t}^{(1:L)}, V_{1:t}^{(1:L)})$. Then*

- (i) (Recurrence) *there exist f, g with $S_t = f(S_{t-1}, x_t)$, $o_t = g(S_t, x_t)$;*
- (ii) (Sufficiency) *o_t depends on $x_{1:t-1}$ only through S_{t-1} ;*
- (iii) (Incrementality) *f appends $2LH_{kv}d_h$ scalars to S_{t-1} and modifies no existing entry; f and g together cost $\Theta(Ld^2 + LtH_{kv}d_h)$ operations, linear in t ;*
- (iv) (Size) $\dim(S_t) = 2tLH_{kv}d_h$.

The trivial recurrence of Remark 5.1 satisfies (i) and (ii) with a smaller state but costs $\Theta(Lt^2d)$ per step.

Proof. For (i), causal masking gives that layer- ℓ outputs at position t depend on positions $< t$ only through $K^{(\ell)}, V^{(\ell)}$; but the map is not componentwise, since for $\ell \geq 2$ the pair $(k_t^{(\ell)}, v_t^{(\ell)})$ is a function of $h_t^{(\ell-1)}$ and hence of S_{t-1} . Define f as the L -fold composition that, for $\ell = 1, \dots, L$, forms $(q, k, v)_t^{(\ell)}$ from $N(h_t^{(\ell-1)})$, appends $(k, v)_t^{(\ell)}$ to $S_{t-1}^{(\ell)}$, and sets $h_t^{(\ell)} = h_t^{(\ell-1)} + \text{Attn}(q_t^{(\ell)}, S_t^{(\ell)}) + \text{MLP}(\cdot)$. This is a finite composition of maps of (S_{t-1}, x_t) , giving (i), and (ii) is immediate from it. For (iii): the projections cost $\Theta(d^2)$ per layer; the readout at layer ℓ is a dot product against t cached keys, costing $\Theta(tH_{kv}d_h)$; no cached entry is modified because $(k_s, v_s)^{(\ell)}$ depends only on $h_s^{(\ell-1)}$ for $s < t$. Summing over L layers gives the bound. (iv) is counting, and the contrast clause is the cost of re-running the forward pass on a length- t prefix. $\square$

Remark 5.3. Clause (iii) is the only one distinguishing the KV cache from storing the raw tokens, and it is what the informal claim omits. Under grouped-query attention the size in (iv) carries an extra factor H_{kv}/H , typically 1/8 to 1/16, which is directly in competition with the capacity question of Theorem 5.12. We also note that “is an RNN” is an abuse: an RNN iterates a *single* map on a *fixed* state space, whereas here the state space grows with t and f is a family of maps with differing domains.

5.2 Linear attention

Theorem 5.4 (Linear attention is a bounded-state recurrence). *Let $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}^{d_k}$ and set $S_0 = 0 \in \mathbb{R}^{d_k \times d_v}$, $z_0 = 0 \in \mathbb{R}^{d_k}$. Then attention with kernel $\exp(\langle q, k \rangle) \mapsto \langle \varphi(q), \varphi(k) \rangle$ satisfies*

$$S_t = S_{t-1} + \varphi(k_t)v_t^\top, \quad z_t = z_{t-1} + \varphi(k_t), \quad o_t = \frac{\varphi(q_t)^\top S_t}{\varphi(q_t)^\top z_t}, \quad (27)$$

a recurrence in the sense of Definition 2.10 with bounded state dimension $d_k d_v + d_k$.

Proof. Unrolling (27) gives $S_t = \sum_{s \leq t} \varphi(k_s)v_s^\top$ and $z_t = \sum_{s \leq t} \varphi(k_s)$, so $o_t = \sum_{s \leq t} \langle \varphi(q_t), \varphi(k_s) \rangle v_s / \sum_{s \leq t} \langle \varphi(q_t), \varphi(k_s) \rangle$, which is causal attention with the stated kernel. Non-negativity of φ ensures the denominator is a sum of non-negative terms, so it neither vanishes by cancellation nor changes sign, and o_t is a convex combination of $\{v_s\}_{s \leq t}$. $\square$

Remark 5.5 (Gating, and a correction). The gated form $S_t = \text{Diag}(a_t)S_{t-1} + \varphi(k_t)v_t^\top$ requires care: decaying S while leaving z_t undecayed drives $o_t \rightarrow 0$, so either z is gated identically or — as in practice — the explicit normaliser is dropped in favour of output normalisation. Further, the structured-masked-attention duality of [15] applies to *scalar* gates $a_t I$: unrolling the vector-gated recurrence gives attention weight $\langle \varphi(q_t) \odot A_{t,s}, \varphi(k_s) \rangle$, which factors as a scalar mask times $\langle \varphi(q_t), \varphi(k_s) \rangle$ if and only if a_t is a multiple of the identity. Vector gating is strictly more general and has no such dual.

5.3 The HMM identity, stated correctly

Theorem 5.6 (Gated dense recurrence and the forward algorithm). *Let $A \in \mathbb{R}^{m \times m}$, $b : \mathcal{O} \rightarrow \mathbb{R}^m$, and define*

$$s_1 = \pi \odot b(o_1), \quad s_t = (A^\top s_{t-1}) \odot b(o_t) \quad (t \geq 2). \quad (28)$$

Then:

- (i) (Algebraic identity; no hypotheses) (28) is *termwise identical to the forward recursion* $\alpha_t(j) = [\sum_i \alpha_{t-1}(i) A_{ij}] B_j(o_t)$ with $b_j(o) = B_j(o)$.
- (ii) (Semantics) if additionally π is a distribution, A is row-stochastic and $B_j(\cdot)$ are emission distributions, then $s_t(j) = \mathbb{P}[o_{1:t}, z_t = j]$ and $\sum_j s_t(j) = \mathbb{P}[o_{1:t}]$.
- (iii) (Viterbi) if $A, b \geq 0$, replacing $A^\top s$ by $(\max_i A_{ij} s(i))_j$ yields $s_t(j) = \max_{z_{1:t-1}} \mathbb{P}[o_{1:t}, z_{1:t-1}, z_t = j]$; recovering the maximising path additionally requires back-pointers.

Proof. (i) Coordinate j of $A^\top s_{t-1}$ is $\sum_i s_{t-1}(i) A_{ij}$, and the Hadamard product multiplies it by $b_j(o_t)$; this is a rearrangement of a matrix-vector product and uses no sign or normalisation hypothesis. (ii) By induction: the base case $s_1(j) = \pi_j B_j(o_1) = \mathbb{P}[o_1, z_1 = j]$ uses the standard convention that π is the law of z_1 ; the step is the law of total probability conditioned on z_{t-1} . (iii) The $(\max, \times)$ semiring distributes over $\mathbb{R}_{\geq 0}$, which is where non-negativity enters and is the only place it is needed; the induction is otherwise identical. $\square$

Remark 5.7 (Which hypotheses do what). Part (i) needs nothing; row-stochasticity buys only the probabilistic reading in (ii); non-negativity is load-bearing only for (iii). Note also the base case: initialising with $s_0 = \pi$ and applying (28) at $t = 1$ applies one spurious transition, and is *not* equivalent to (28) as stated.

5.4 The bridge does not exist

It is tempting to read Theorem 5.6 as a constrained special case of gated linear attention (Theorem 5.4), i.e. to claim that making a gated linear RNN’s state non-negative and its gate stochastic turns it into an HMM. It does not, and the obstruction is structural rather than technical.

Theorem 5.8 (No embedding of the forward algorithm into gated linear attention). *Let $\Phi(S) = \text{Diag}(a)S + C$ be one step of the gated linear-attention recurrence with any gate a and any input-determined C , and let $\Psi_o(s) = \text{Diag}(b(o))A^\top s$ be one step of (28). For a generic HMM (A, B) with A strictly positive there is no injective linear $\iota : \mathbb{R}^m \rightarrow \mathbb{R}^{d_k \times d_v}$ and no choice of gates and inputs satisfying $\Phi \circ \iota = \iota \circ \Psi_o$ for all o .*

Proof. Two independent obstructions.

(a) *Support monotonicity.* For any S, S' we have $\Phi(S) - \Phi(S') = \text{Diag}(a)(S - S')$, since C does not depend on the state. Hence the set of rows on which two trajectories differ is non-increasing along the recurrence, for every gate and every input. By contrast, taking $s - s' = e_r$ gives $\Psi_o(s) - \Psi_o(s') = (b_j(o)A_{rj})_j$, whose support is $\{j : A_{rj} > 0, b_j(o) > 0\}$ — all of $[m]$ for strictly positive A and b . Support strictly grows, so no coordinatewise identification of the HMM state with rows of S can be equivariant.

(b) *Simultaneous diagonalisability.* Allowing an arbitrary injective linear ι , the conjugacy $\Phi \circ \iota = \iota \circ \Psi_o$ restricted to $\text{im } \iota$ states that every Ψ_o is diagonal in the single basis $\iota(e_1), \dots, \iota(e_m)$. Simultaneously diagonalisable matrices commute, so this forces $[\Psi_o, \Psi_{o'}] = 0$ for all pairs o, o' , which fails for generic (A, B) . Hence no such ι exists. $\square$

Proposition 5.9 (The common generalisation). *Call $s_t = M(u_t)s_{t-1} + c(u_t)$, $s_t \in \mathbb{R}^N$, an input-controlled affine recurrence. Gated linear attention is the case $M(u_t) = \text{Diag}(a_t) \otimes I_{d_v}$ — block diagonal, no mixing across the d_k index — with $c(u_t) = \text{vec}(\varphi(k_t)v_t^\top)$; the forward algorithm is the case $M(o_t) = \text{Diag}(b(o_t))A^\top$ dense with $c \equiv 0$. These are two incomparable cases, not a case and its restriction.*

Remark 5.10 (Consequence for the complaint). A gate *is* a diagonal matrix; constraining its entries to be non-negative and to sum to one yields a diagonal stochastic matrix, that is, the identity — an HMM whose hidden state never moves. The transition matrix must be *added* to the architecture, not recovered by constraining something already present. Diagonality is not incidental: it is precisely what lets [14, 15] evaluate the recurrence as an associative scan. Architectures whose state transition is dense, such as the delta rule’s $I - \beta_t k_t k_t^\top$ [16], are the ones for which an HMM reading is even a candidate. The informal thesis — “constrain a gated linear RNN and you get HMM-like decoding” — is therefore false as stated, and the defensible version is that HMM-like decoding requires giving up the diagonal structure that makes these models fast.

5.5 What unbounded state buys

Definition 5.11 (m -bit statefulness). A sequence model is m -bit stateful if for every t its state S_t takes values in a set of cardinality at most 2^m , and its emissions after time t are a deterministic function of S_t and subsequent inputs.

The finiteness in Definition 5.11 is essential and is exactly what the informal version of the next result omits: without it the bound is simply false, since $s_t = s_{t-1}/2 + x_t/2$ stores an unbounded binary expansion in one real number and can be decoded indefinitely by $x \mapsto \lfloor 2s \rfloor$, $s \mapsto 2s - \lfloor 2s \rfloor$.

Theorem 5.12 (Recall lower bound). *Let a model be m -bit stateful and suppose that for every $c \in \mathcal{V}^n$, when presented with a context containing c at a designated location followed by a retrieval cue, it emits c exactly. Then $m \geq n \log_2 V$. If instead it succeeds with probability $1 - \delta$ over a uniform c , then $m \geq (1 - \delta) n \log_2 V - 1$.*

Proof. For the exact case, let $S(c)$ be the state reached after the context containing c and before the cue. Emissions after that point are a function of S and the cue alone, so $c \mapsto S(c)$ must be injective on $\mathcal{V}^n$; hence

$2^m \geq V^n$. For the probabilistic case, let C be uniform on $\mathcal{V}^n$ and $\hat{C}$ the emitted block, a function of S . By Fano’s inequality $H(C | S) \leq 1 + \delta n \log_2 V$, while $H(C | S) \geq H(C) - H(S) \geq n \log_2 V - m$; rearranging gives the claim. $\square$

Remark 5.13. The order of quantifiers matters: the block contents must be fixed before the state forms, and if the retrieval *location* is also adversarial it must be chosen afterwards, in which case the bound becomes $T \log_2 V$ for a length- T context. The converse — that a KV-cache transformer achieves recall — is not a corollary of this lower bound but a separate constructive fact [18, 17]. Finally, the Fano form predicts *graceful* degradation past the capacity threshold rather than a sharp cliff, which is the prediction our window-attention probe should be read against.

6 Experiments

6.1 Setup, and an honest account of scale

Our experimental environment cannot reach `huggingface.co`: the network policy denies the connection, as it does `download.pytorch.org`. Public checkpoints were therefore unavailable and every measurement below is on a model we trained ourselves. That model satisfies Definition 2.1 exactly — pre-norm residual blocks, RMSNorm, an unembedding — with $L = 6$, $d = 128$, $V = 2048$, 1.71M parameters, trained on a 53k-token corpus with a held-out split and early stopping.

We state the consequences plainly rather than in a footnote. The model is *weak*: validation perplexity is 431 against a vocabulary of 2048, i.e. about $4.8\times$ better than uniform. Two specific distortions follow. First, diffuse output distributions overlap more, so measured acceptance rates $\alpha_\ell = 1 - \text{TV}(p_\ell, p_L)$ are *inflated* relative to a well-trained model. Second, at $L = 6$ the shallowest available exit is $\rho = 0.167$, whereas a 28-layer model offers far finer granularity exactly in the regime where early exit is attractive. Quantitative claims below should therefore be read as indicative; the qualitative ones (Sections 6.2 and 6.4) do not depend on scale, and the two errors we report finding are architecture-independent.

6.2 Exactness is confirmed at the sampling-noise floor

Theorem 3.1 predicts that speculative sampling with *any* draft returns the target distribution exactly. Testing this against zero would be meaningless, since any finite sample has positive total variation from p_L ; the correct reference is the noise floor, namely TV between p_L and an equal number of direct draws from it. Table 1 reports both for 4×10^5 samples.

layer	TV(spec, p_L)	TV(direct, p_L)	ratio	$1 - \text{TV}(p_\ell, p_L)$	observed accept
1	0.0195	0.0198	0.984	0.4635	0.4634
3	0.0202	0.0196	1.033	0.5831	0.5833
5	0.0186	0.0193	0.965	0.7988	0.7978

Table 1: Speculative sampling is indistinguishable from direct sampling (ratio ≈ 1), and the observed acceptance rate matches the predicted $1 - \text{TV}(p_\ell, p_L)$ to within 10^{-3} . These rates are measured at a single position and so differ from the corpus-averaged α_ℓ quoted elsewhere; at $\ell = 1$ the single-position value is 0.4635 against a corpus average of 0.3274.

This is the result that licenses Corollary 3.24, since the shortlisted draft assigns zero probability to most of the vocabulary and would be invalid under any support hypothesis.

6.3 The acceptance bound is directionally right and numerically useless

Acceptance rises monotonically with depth exactly as tail energy falls: $\alpha_\ell = 0.327, 0.383, 0.424, 0.520, 0.693, 1.000$ against $T_\ell = 20.2, 17.8, 15.4, 12.6, 8.1, 0$. The shape predicted by Theorem 3.7 is therefore present.

The bound itself is not usable. At $\ell = 5$, where $\alpha_5 = 0.693$, feeding the bound the *measured* logit spread gives $e^{-4.21} = 1.5 \times 10^{-2}$, already a factor of 46 too small; the Cauchy–Schwarz step then inflates the spread by a further $2.5\times$, landing at $\approx 10^{-5}$. Remark 3.9 anticipated looseness and attributed it primarily to the D_U step. That attribution was wrong: the softmax stage alone is fatal, and Cauchy–Schwarz merely compounds it. A useful bound needs a different proof strategy, not a tighter constant.

6.4 A cost-model error, caught by a sanity check

The most consequential empirical finding is an error in our own accounting. An earlier version of Proposition 3.16 charged drafting and verification but omitted the full-depth pass at the bonus position $t + \gamma$. Evaluated at $\rho = 1$ it returned $S = 1.735$, which is impossible: at $\rho = 1$ the draft *is* the target, so the procedure degenerates to ordinary autoregressive decoding and $S = 1$ is forced. Restoring the omitted pass gives exactly $S = 1.000$ (Remark 3.17). We recommend the $\rho \rightarrow 1$ limit as a standing check on any speculative-decoding cost model; it is cheap and it caught an error that survived both derivation and adversarial review.

The same experiment overturned our claim about cache reuse. Measured at $w = 1$, reuse is *worse* at every depth (2.25 against 1.50 at $\rho = 0.25$, $\gamma = 2$), because splitting verification across two depths pays for the lower blocks’ weight streaming twice; at $w = 0$ it helps, 0.929 against 0.667. This is Proposition 3.18, and Remark 3.19 retracts the earlier claim.

6.5 The head cost decides whether self-drafting pays

With the corrected model, the best attainable memory-bound speedup on our model is $S = 1.026$, at $\ell = 1$, $\gamma = 1$, with head fraction $u = 0.153$. Setting $u = 0.26$, the value for Qwen3-0.6B, *no* early-exit layer exceeds $S = 1$: the optimum retreats to the degenerate $\rho = 1$. The margin is entirely determined by the unembedding cost, exactly as Corollary 3.22 predicts, and a 2.6% gain on a model whose acceptance rates are already inflated is not a result we would build on.

One further asymmetry deserves recording. At $\ell = 1$ the sampling-mode acceptance rate is $\alpha_1 = 0.327$ while top-1 agreement between p_1 and p_L is only 0.052. These govern different deployment modes — the former speculative sampling at temperature 1, the latter greedy decoding — and the gap of more than $6\times$ means an acceptance rate that looks tolerable for sampled decoding is close to useless for greedy decoding. Reported acceptance rates should always name the mode.

6.6 Early-exit training rescues thesis 1, marginally

Remark 3.26 located the obstruction to self-drafting in the training objective: intermediate layers are never asked to be decodable. We tested that claim directly by fine-tuning the same model for 600 steps with an auxiliary early-exit loss, $\mathcal{L} = \text{CE}_L + \frac{1}{2} \text{mean}_{\ell < L} \text{CE}_\ell$ through the shared head, and re-measuring everything in Sections 6.2–6.4. Table 2 gives the per-layer effect.

Three observations. First, exactness is unaffected (TV ratios 1.015, 0.930, 0.998), as Theorem 3.1 requires of any draft. Second, the sampling-mode/greedy-mode gap of Section 6.4 largely closes: at $\ell = 1$ top-1 agreement rises sixfold while α_1 rises by half. Third, and decisively for the thesis, the corrected memory-bound speedup becomes

head fraction	untuned	trained, 600 steps ($\Delta\text{val} + 0.08$)	trained, early-stopped ($\Delta\text{val} - 0.03$)
$u = 0.153$ (this model)	1.026	1.146	1.097
$u = 0.260$ (Qwen3-0.6B)	1.000	1.072	1.044

ℓ	α_ℓ before	after	top-1 before	after	T_ℓ before	after
1	0.327	0.483	0.052	0.312	20.2	15.3
2	0.383	0.586	0.115	0.354	17.8	13.0
3	0.424	0.677	0.167	0.521	15.4	10.8
4	0.520	0.751	0.271	0.745	12.6	8.6
5	0.693	0.844	0.552	0.839	8.1	5.8

Table 2: Effect of 600 steps of auxiliary early-exit loss. Acceptance rises at every layer, greedy top-1 agreement rises far more, and the residual tail energy falls at every layer — the auxiliary loss moves intermediate states toward the final one, which is the mechanism by which Theorem 3.7 says acceptance is governed. Final-layer validation loss rose by 0.08.

At Qwen’s head fraction the optimum was previously the degenerate $\rho = 1$; it is now a genuine early exit. The early-stopped run, which keeps the state of best final-layer validation loss (step 100, an *improvement* of 0.03), still clears 1 at 1.044; the extra three points of the full run are bought with 0.08 of final-layer loss. Thesis 1 therefore survives as an engineering claim on this model at zero quality cost — by 4%, after training, on a model whose acceptance rates are inflated by its weakness. We would not build on a margin of that size, but the sign is the one Remark 3.26 predicted, and the mechanism (falling T_ℓ) is the one the theory names.

The within-round yield decay of the next section is *stronger* after training: measured $\mathbb{E}[Y] = 1.435$ against a geometric 1.882 at $\bar{\alpha} = 0.483$, with $\mathbb{P}[A_1] = 0.340$ already below $\bar{\alpha}$. Acceptance measured on held-out text overstates acceptance during decoding, because decoding visits the model’s own continuations; the gap widens as the draft improves on the text distribution.

6.7 Yield: acceptance decays faster than geometric

Running 200 real speculative rounds at $\ell = 1$, $\gamma = 4$ gives measured $\mathbb{E}[Y] = 1.320$ against 1.481 from the geometric formula of Corollary 3.13 evaluated at $\bar{\alpha} = 0.327$. The measured prefix probabilities are $\mathbb{P}[A_j] = 0.250, 0.055, 0.010, 0.005$ against $\bar{\alpha}^j = 0.327, 0.107, 0.035, 0.011$ under independence: acceptance decays *faster* than geometric within a round.

This contradicts the remark following Corollary 3.13, which argued from Jensen’s inequality that cross-context heterogeneity makes the geometric formula a conservative under-estimate. That argument concerns variation *between* contexts; within a round the dominant effect runs the other way, because each accepted draft moves the model onto its own continuation, where the shallow-layer draft agrees less well. Practitioners estimating speedup from a mean acceptance rate will therefore *over*-state yield, by 12% here.

We note that the printed identity $\mathbb{E}[Y] = 1 + \sum_j \mathbb{P}[A_j]$ is a bookkeeping check only: both sides are computed from the same run, so it verifies our accounting rather than testing Proposition 3.12, which is in any case a definitional identity.

6.8 The HMM identity and the impossibility of the bridge

Theorem 5.6 is verified against brute-force enumeration over all M^T hidden paths, per (t, j) rather than on aggregates of the final state, for 20 random HMMs. A negative control confirms the hypothesis split: with a signed transition matrix the sum-product identity still holds exactly (it is pure algebra) while the max-product reading breaks, since $(\max, \times)$ distributes only over $\mathbb{R}_{\geq 0}$.

Theorem 5.8 is confirmed directly. Under a gated linear-attention update the set of rows on which two trajectories differ stays at $\{1\}$; under the forward algorithm a single differing coordinate spreads to all of $[m]$ in one step. The commutator of two observation-conditioned transition operators is $\|\Psi_0\Psi_1 - \Psi_1\Psi_0\|_{\max} = 4.5 \times 10^{-2}$, so they are not simultaneously diagonalisable. Both obstructions hold numerically.

6.9 Interface bandwidth, and an artefact that would have flattered us

Theorem 4.10 predicts that a streaming step machine solves $\text{PROD}_{N,T}$ iff its interface carries at least $\log_2(N!)$ bits. For $N = 5$ that threshold is 6.91 bits. Table 3 sweeps the interface width at $T = 4$.

interface (bits)	3	4	5	6	7	8	15	$\mathbb{R}^{96}$
accuracy	0.359	0.545	0.614	0.786	0.852	0.917	1.000	1.000

Table 3: Accuracy against interface width, $N = 5$, $T = 4$. The information-theoretic threshold is $\log_2 5! = 6.91$ bits. No width below it reaches success, accuracy is monotone in width, and the success threshold is crossed at 8 bits.

The bound is respected — no sub-threshold width succeeds — and degradation below it is smooth rather than cliff-like, again matching the Fano reading. Crossing at 8 rather than 7 bits indicates roughly one bit of coding overhead: gradient descent does not find the dense Lehmer code that saturates the entropy bound exactly. Consistently, an interface of 15 bits — the “natural” code of five positions at three bits each — succeeds outright.

We report a negative methodological result alongside it, because it is the one artefact in this paper that would have *supported* our thesis. Our first sweep used $T = 12$ and found the discrete interface scoring below 0.06 at every width from 3 to 10 bits while the continuous interface scored 1.000. Read naively that is a textbook confirmation: discrete channel bad, continuous channel good, exactly as the informal bandwidth argument predicts. It is not a bandwidth effect at all. A bound binding at 6.91 bits cannot explain failure at 10 bits, and the diagnostic is decisive: at $T = 12$ even a 20-bit interface scores 0.037, while at $T = 4$ a 7-bit interface scores 0.872. The binding constraint at $T = 12$ is credit assignment through twelve stacked hard quantisations, whose straight-through gradients degrade with the number of bottlenecks on the path. The two failure modes are separable because optimisation failure is monotone in T while bandwidth failure is monotone in b .

This is the third occasion in this work on which an optimisation artefact imitated a capacity result, after the undertrained sweep and the one-hot parameterisation of Section 6.12. It is the only one whose naive reading favoured our own hypothesis, which is precisely why we pursued it. We suggest the general check: before reading a discrete-bottleneck failure as evidence of a capacity bound, verify that the failure is monotone in the bottleneck width and absent at widths well above the claimed threshold.

6.10 Discretisation as error correction: the learned case is unresolved

Proposition 4.15 settles the idealised version of Open Problem 4.14: no lock-in, $k = 1$ optimal at sufficient rate, a binary switch at insufficient rate. We attempted the learned version — a trained write, a trained 16-bit codebook, a trained read, 8 bits carried through 64 steps of identity-plus-noise — and report it as inconclusive. At $\sigma = 0$, where projection costs nothing, accuracy was 1.00 with no projection, 0.75 with four, and chance with sixteen or sixty-four. Monotone in the number of projections at zero noise is an optimisation signature, not a capacity or error-correction one, and it is the fourth time in this work an optimisation failure has presented as a result. The script now refuses to report unless the $(\sigma = 0, k = 1)$ cell is itself learnable. The learned-codebook question stays open.

6.11 Theorem 5.8 has bite, and where

Theorem 5.8 is an exact-embedding result and says nothing about approximation. If a diagonally-gated recurrence approximates HMM predictions to within noise, the theorem is true and practically empty. We trained input-controlled affine recurrences (Proposition 5.9) with diagonal or dense action on sequences from a known HMM and scored each by its excess next-symbol NLL over the forward algorithm, which is Bayes-optimal.

The first HMM we tried nearly misled us in the direction *against* our own result. On a random HMM with sharply peaked rows, a single diagonal layer at matched state size reached 0.2% of the available headroom, the same as a dense layer, and the script printed a disproof. But that HMM’s mean belief entropy is near zero: the posterior collapses after a symbol or two, no mixing is ever needed, and a diagonal recurrence matching a dense one is expected. The script now measures belief entropy and refuses to rule unless the hidden state must actually be tracked. Table 4 is the HMM that can decide it: transitions are a noisy cyclic permutation, emissions are mildly informative, and the belief *rotates* — the transition operator has complex eigenvalues, exactly the case the proof of Theorem 5.8(b) singles out.

	dense-1	diag-1	diag-2	diag-3
$d = m = 4$	5.8%	55.8%	53.1%	46.6%
$d = 16$	2.9%	38.1%	27.5%	25.2%

Table 4: Excess NLL as a percentage of headroom (uniform minus optimal) on the cyclic HMM, $m = 4$ states, belief entropy 0.923 of $\log 4 = 1.386$. A single dense layer captures 94% of what is learnable; a single diagonal layer captures 44%. Neither two extra layers nor fourfold width closes the gap.

The exact non-embeddability therefore translates into a tenfold gap in excess loss, and it appears precisely where the theory locates it: when the belief must be carried through a mixing transition rather than re-identified from recent symbols. This is the empirical content of Remark 5.10. Diagonal gating is not a harmless parameterisation choice on such sequences; it is the thing that prevents the model from tracking the state, and the remedy is a dense transition, not depth.

6.12 Capacity: the bound is never violated

Theorem 5.12 predicts that a b -bit interface retains n binary values iff $b \geq n$. Table 5 sweeps both.

$b \backslash n$	2	3	4	5
1	0.51	0.67	0.63	0.60
2	1.00	0.83	0.75	0.71
3	1.00	1.00	0.76	0.71
4	1.00	1.00	1.00	0.81
5	1.00	1.00	0.87	1.00

Table 5: Recall accuracy by interface width b and number of retained values n . The diagonal $b = n$ is bold. No cell with $b < n$ reaches success, which is the only direction in which a lower bound can fail.

Nineteen of twenty cells match the prediction. The exception, $b = 5$, $n = 4$ at 0.87, is optimiser noise rather than a capacity effect: $b = 4$, $n = 4$ reaches 1.00 in the same sweep and five bits strictly contain four. Degradation below the diagonal is graceful (0.83, 0.76, 0.71 at $b = n - 1$), matching the Fano form of Theorem 5.12 and contradicting the sharp cliff we originally pre-registered; that prediction is withdrawn.

Two optimisation artefacts had to be cleared first, and both initially looked like capacity results. An early sweep at 800 training steps produced seven failures *with sufficient capacity*, including $b = 4$, $n = 2$, where 16 symbols must hold 2 bits; no capacity argument can explain that, since $b = 3$, $n = 2$ succeeded in the same sweep, and at 3000 steps the cell reaches 1.000 on every seed. Separately, parameterising the interface as one symbol from a codebook of size 2^b — the same b bits — never cleared threshold at $n \geq 3$ even at 7000 steps with two seeds, because it requires the optimiser to *discover* an injective code through a biased straight-through gradient. Writing the interface as b independent binary units, so that the natural solution “unit i holds value i ” is directly representable, produces the table above. Capacity is identical in both cases; only trainability differs.

7 Iterating the architecture: a theory-guided search

The three axes are not a taxonomy but a design space, and the results above are predictions about which moves in it should pay. We tested that by running a small architecture search in which every claim of Sections 3–5 is a switch next to its naive alternative, under a protocol fixed before the first candidate was scored.

7.1 Architecture, benchmark, protocol

Architecture. A four-block residual model of width 64 in which each block’s sequence mixer is one of `diag` (input-controlled diagonal gating, the action of gated linear attention), `dense` (a static dense transition with a per-token multiplicative gate, the shape of Theorem 5.6), or `attn` (causal softmax attention). The *time* axis adds r latent loops over thought slots that either overwrite or append (Corollary 4.5) with optional codebook projection every k loops (Proposition 4.15); the *depth* axis adds an early-exit auxiliary loss λ on the shared head (Section 6.6).

Benchmark. One procedurally generated task per axis on a shared vocabulary: `RECALL` (six key/value pairs, then a query; state capacity), `TRACK` (cyclic-HMM next-symbol prediction scored as the fraction of headroom captured against the forward algorithm; state mixing), and `COMPOSE` (eight generators of S_5 to their product; serial time). Depth is the corrected-cost early-exit speedup S of Proposition 3.16 at an imposed head fraction $u = 0.26$, measured on all three tasks. Fitness is $F = \text{state} + \text{time} + \text{depth} \in [0, 3]$ with depth credit gated by task quality, since an untrained model has every layer near-uniform and would otherwise score full depth credit for predicting nothing.

Protocol. Parameters $\leq 1.25\times$ and compute $\leq 3.5\times$ a baseline reference, enforced before any compute is spent. Two seeds per candidate, three for the baseline, held-out evaluation on fresh samples. The noise band τ is the baseline’s across-seed standard deviation of F , fixed before any candidate runs; a candidate replaces the incumbent iff it beats it by more than τ . Every candidate is tagged with the axis it mutates and the result that motivates it. The first three candidates are the three theory-predicted moves, each applied to the naive baseline in isolation, so each axis’s contribution is measured before any are combined.

7.2 Search v1: thirteen rounds, one accepted move, and why

On the naive baseline ($F = 0.813 \pm 0.097$, so $\tau = 0.097$) the early-exit loss was accepted at $+0.302$, moving S from 1.12 to 1.67 while leaving every task score unchanged — the effect of Section 6.6 reproduced on a different model and different tasks. Dense mixing moved `TRACK` in the predicted direction ($0.51 \rightarrow 0.57$) but landed at $+0.084$, under the band. Two appended loops were *worse* by 0.136 at three times the compute.

Ten further rounds — stacking, more λ , two attention layers, deeper exits only, all-dense, two learning rates, a matched append-versus-overwrite pair, projection every step — produced nothing above the band. The diagnosis, reached by diagnostics rather than assumed, was that two axes were pinned by the benchmark: `RECALL` sat at 0.33–0.36 for *every* layout including four attention layers trained on it alone with four times the budget, and `COMPOSE` reached 0.27 at best. A localiser sweeping the number of pairs found the pipeline sound (a one-pair copy trains to 1.000) and the failure beginning at two pairs, the first point a key match is required: two-hop retrieval does not form in this model at this budget. Architecture cannot register on tasks no architecture can learn. Two results from v1 stand on their own: the matched loop test went *against* Corollary 4.5 (overwrite beat append by 0.104 at $r = 1$, both worse than no loop), and projecting every step on a noise-free loop changed nothing (0.996 against 0.998), as Proposition 4.15 says it should not.

7.3 Search v2: the same rule on a learnable encoding

v2 changed the `RECALL` encoding to one token per pair — a one-hop content match with identical semantics — and nothing else, after a gate showed all-attention now climbs ($0.27 \rightarrow 0.42 \rightarrow 0.66$ and rising) where

diagonal state does not (0.375). The v2 baseline scored 0.934 ± 0.015 , so $\tau_2 = 0.015$. Table 6 gives the pre-committed rounds.

round	mutation	F	Δ	verdict
14	naive baseline (3 seeds)	0.934 ± 0.015	—	incumbent
15	depth: $\lambda = 0.5$	1.236 ± 0.006	+0.302	accept
16	state: dense, dense, attn, dense	1.073 ± 0.116	+0.139	accept
17	time: append, $r = 1$	0.699 ± 0.079	−0.235	worse
18	time: overwrite, $r = 1$	0.728 ± 0.055	−0.206	worse
19	depth + state	1.513 ± 0.224	+0.579	accept

Table 6: Search v2 against the naive baseline, $\tau_2 = 0.015$. Depth and state both register and combine additively; both loop variants are worse than no loop, and overwrite again beats append (by 0.029, above the band).

Three things follow. The depth move is the most robust prediction in this paper: +0.302 in both searches, on both encodings, with the task scores untouched. Dense mixing pays exactly where Theorem 5.8 and Section 6.11 say it should — TRACK rose from 0.55 to 0.63 alone and to 0.68 combined — while RECALL did not move with or without an attention layer in the three-task mix, so the recall layer may be dead weight; round 20 tests that. And latent loops have hurt at every budget and encoding tried, with the append-versus-overwrite comparison going against Corollary 4.5 twice. The corollary is a capacity statement about long horizons; at $r = 1$ on a model that cannot yet use one loop, capacity is not what binds, and we record the loss rather than explain it away. Round 20 dropped the attention layer (all-dense with the depth loss) and tied the incumbent on the search seeds (+0.012, under τ_2).

7.4 The claim, on seeds the search never saw

Table 7 re-trains each v2 candidate on three fresh seeds.

config	F (held-out)	state	time	depth	TRACK
naive baseline	0.873 ± 0.032	0.450	0.216	0.207	0.557
+ depth	1.192 ± 0.037	0.416	0.180	0.596	0.499
+ state	1.074 ± 0.100	0.502	0.287	0.285	0.602
+ depth + state (round 19)	1.560 ± 0.056	0.482	0.298	0.780	0.644
all-dense + depth (round 20)	1.460 ± 0.075	0.481	0.249	0.730	0.654

Table 7: Held-out evaluation on seeds 5–7. The round-19 configuration is 79% above the naive baseline; its ± 0.224 search-seed spread tightens to ± 0.056 . Round 20’s search-seed tie reverses to a 0.100 deficit: the attention layer was not dead weight.

The search therefore ends where Sections 3 and 5 point: a dense-mixing recurrence with one exact-recall layer, trained so its intermediate layers are decodable, and no latent loops. Depth contributes +0.319 and state +0.201 on held-out seeds, combining to +0.687. The time axis, as tested in rounds 1–20, contributed nothing and cost something; the next twenty rounds show why that verdict was premature.

7.5 Rounds 21–40: an autonomous continuation

A second twenty rounds ran unattended under a stricter rule — three seeds per candidate, τ the incumbent’s own spread capped at 0.20, and a held-out re-training on fresh seeds before any promotion. One candidate was promoted: learning rate 4×10^{-3} (round 21), taking held-out F from 1.560 to 1.806. It is the largest single

move in this paper and it is a recipe, not an architecture; the first search saw it at $+0.057$ under a wide τ and could not accept it. Nothing else cleared the band, including every near-miss stacked ($+0.187$ at round 40 against $\tau = 0.20$).

Two findings matter more than the promotion. First, *the time axis had never been measured*. Round 32 inserted placeholder slots with zero loops and lost by 0.537 — the same collapse as every loop configuration before it. A dense recurrence with a spectrally normalised transition and gates below one contracts over four steps of constant input, so filler tokens erase its memory; attention would not care. Holding recurrent state at placeholder positions removed the penalty exactly (round 36 matches the incumbent), and rounds 37–38 became the first fair loop tests: both lose by about 0.48 , and append ties overwrite ($+0.018$). Corollary 4.5’s direction therefore survives the one honest test it has had; loops as a mechanism, on this benchmark at this budget, do not help. The earlier “overwrite beats append” results were layout artefacts and are withdrawn.

Second, the state axis found its mechanism in a losing round. RECALL sat at 0.25 – 0.36 under every layout for thirty-four rounds; round 34, with attention at both ends, lifted it to 0.84 while TRACK and COMPOSE collapsed, and a follow-up with attention first confirmed the pattern (0.86 against 0.11). Content lookup needs attention on raw token embeddings, before the recurrence’s normalised mixing erases key identity; composition needs dense mixing on those same raw tokens. The first layer is contested, and a four-block model has one. This is a structural limit of the architecture, not a failure of the search, and it says what a fifth block or a parallel first layer would be for.

Table 8 re-trains the second search’s candidates on seeds 5–7. The promotion holds ($+0.246$ over round 19); the placeholder control sits inside the incumbent’s spread, so holding recurrent state at filler positions costs nothing; and the recall mechanism replicates, 0.976 – 0.978 with attention on raw tokens against 0.28 – 0.34 for every other layout. One row is a surprise. Round 39 ($\lambda = 0.75$, attention last), rejected at $+0.127$ under $\tau = 0.20$, beats the incumbent on every fresh seed ($+0.163$; its worst seed, 1.885 , is above the incumbent’s mean). The incumbent’s own spread, ± 0.215 and the widest in the table, set the τ that excluded it. We do not promote it on this evidence — these seeds are confirmation data, and selecting on them is what the protocol forbids — but it is the pre-registered first move of any continuation: both configurations on seeds 8–10 under the same rule. That test ran: incumbent 1.801 ± 0.271 , round 39 1.908 ± 0.382 , $\Delta = +0.107$ against $\tau = 0.20$. Round 39 wins two seeds clearly and ties the third, where both collapse; it is not promoted, and the incumbent stands. Across six fresh seeds it leads by $+0.135$, a lean the protocol cannot resolve at three seeds per test. More seeds would settle it; more rounds would not.

config	F (held-out)	depth	RECALL	TRACK	COMPOSE
naive baseline	0.873 ± 0.032	0.207	0.343	0.557	0.216
round 19 (first incumbent)	1.560 ± 0.056	0.780	0.320	0.644	0.298
round 21, $lr\ 4 \times 10^{-3}$ (incumbent)	1.806 ± 0.215	0.903	0.277	0.859	0.335
round 36, placeholder control	1.718 ± 0.094	0.859	0.286	0.851	0.290
round 39, $\lambda = 0.75 + \text{attn last}$	1.969 ± 0.074	0.981	0.311	0.877	0.395
round 34, attn at both ends	1.631 ± 0.113	0.816	0.976	0.418	0.119
round 41, attn first	1.657 ± 0.028	0.829	0.978	0.435	0.122

Table 8: Held-out evaluation of rounds 21–41 on seeds 5–7, never seen by either search. Attention on raw tokens triples RECALL and halves COMPOSE; no four-block layout gets both. Round 39 is the one configuration the rule rejected that fresh seeds prefer.

8 A unified object, and how to falsify these claims

8.1 What the three results describe together

Placed side by side, the surviving results do not describe three patches but one design, with each axis constrained by a different theorem:

- **State.** A learned, bounded, gated recurrence (Theorem 5.4) — but with a *dense* transition if an HMM-like reading is wanted (Theorem 5.8), and with enough exact-recall capacity to clear Theorem 5.12.
- **Time.** Serial steps that need not emit a token (Theorem 4.7), but which *accumulate* rather than overwrite (Corollary 4.5), with projection onto a learned codebook every k steps (Open Problem 4.14).
- **Depth.** Early exit governed by tail energy (Theorem 3.7), supplying both an adaptive halting signal and, on the tokens that do not halt, a free speculative draft (Theorem 3.20).

The depth axis serves two purposes at once: the signal that says “this token is easy, emit it now” is the signal that produces a draft for the tokens that are not. Adaptive computation time [20] and self-speculation are the same mechanism read in opposite directions. We prove nothing about the combination and regard the experiments below as prior to any such attempt.

8.2 Protocol

Each experiment targets one theorem and carries a kill criterion fixed in advance. Models are $\leq 1\text{B}$ parameters; two experiments require no training.

E1: acceptance profile (Theorems 3.7, 3.20). For every layer ℓ of Qwen3-0.6B, measure $\alpha_\ell = 1 - \text{TV}(p_\ell, p_L)$, tail energy T_ℓ , and the bound (16), then evaluate the speedup surface with the corrected cost (20) including the head fraction $u \approx 0.26$. Report the looseness of (16) decomposed into its Cauchy–Schwarz and softmax stages, and compare against Proposition 3.10 with P fitted to the post-norm tail. **Kill:** $\max_{\ell, \gamma} S \leq 1$ both untuned and after an early-exit auxiliary-loss adaptation.

E2: bandwidth (Theorem 4.10). Train $\sim 10\text{M}$ -parameter models from scratch on $\text{PROD}_{N,T}$ under a *streaming* read schedule, which Proposition 4.12 shows must be enforced rather than assumed: the re-read escape is otherwise open, and because composition is associative, recomputation costs only $\log_2 T$ depth, so a 4-layer model needs $T \gtrsim 2^8$ before depth alone closes it. Sweep the interface width against $\log_2(N!)$ and check the predicted threshold at $N = 9$ (Corollary 4.11). **Kill:** accuracy does not fall at $b < \log_2(N!)$ once the read schedule is enforced.

E3: the HMM identity and the missing bridge (Theorems 5.6, 5.8). Verify Theorem 5.6 numerically against brute-force path enumeration, per state and per timestep, with a negative control that drops non-negativity. Then verify Theorem 5.8 directly: check that row-support is non-increasing along a gated linear-attention recurrence and strictly increasing along the forward algorithm, and that the Ψ_o fail to commute. **Kill:** the identity fails, or the support and commutator tests come out the other way.

E4: capacity (Theorem 5.12). Restrict Qwen3-0.6B’s attention to a sliding window of w tokens as a bounded-state proxy and locate the associative-recall break-point as a function of w . Note that the Fano form of Theorem 5.12 predicts *graceful* degradation past threshold rather than a cliff. **Kill:** the break-point does not move with w , which would falsify the proxy rather than the theorem.

8.3 Limitations

The cost model of Definition 2.4 ignores attention’s quadratic term and KV traffic. Theorem 4.1 is a statement about classes and does not constrain any individual trained model (Remark 4.2). Theorem 4.7 and Theorem 4.1 assume incompatible precision regimes. Theorem 4.10 requires a streaming read schedule that real architectures do not obey, and Remark 4.13 explains why removing that hypothesis is not merely difficult but would separate TC^0 from NC^1 .

The experiments of Section 6 are limited by scale and by environment. Our network policy denied access to public checkpoints, so all measurements are on a 1.71M-parameter model with validation perplexity 431; its diffuse output distributions inflate acceptance rates, and its six layers give coarse control over the exit depth ρ . The synthetic experiments are limited instead by optimisation: three separate artefacts in them initially imitated capacity results, and only one direction of failure — success below a claimed threshold — can refute a lower bound. We therefore report violation counts separately from match rates throughout. Nothing here establishes how these effects scale, and the quantitative claim most likely to move with scale is the early-exit speedup, since acceptance at small ρ should improve with depth.

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